KENYA MEDICAL TRAINING COLLEGE

FACULTY OF CLINICAL SCIENCES

DEPARTMENT OF CLINICAL MEDICINE

DIPLOMA IN CLINICAL MEDICINE & SURGERY

YEAR 2, END OF SEMESTER 1 EXAMINATION

PAPER: SURGERY I

DATE: 18th March, 2021 **TIME: 2 HOURS (08.00 A.M – 10:00 A.M)**

INSTRUCTIONS

1. This paper consists of Four (4) sections:
   - Section 1 (10 Multiple Choice Questions)
   - Section 2 (10 True False Questions)
   - Section 3 (4 Short Answer Questions)
   - Section 4 (1 Long Answer Question)
2. Attempt **ALL** Questions.
3. Write your **REGISTRATION NUMBER** on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination.
5. Ensure you sign the examination register provided.
6. Any examination malpractices will be handled as per the College Examination Policy.

REGISTRATION NUMBER.....

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SECTION 1: MULTIPLE CHOICE QUESTIONS - (20 MARKS)

1. Which of the following statements about homeostatis is false?
   a) it is defined as the stable state of the normal body
   b) emergency surgery should cause little disturbance to homeostatsis
   c) the central nervous system, heart, lungs, kidneys and spleen are the essential organs that maintain normal homeostasis
   d) elective surgery should cause little disturbance to homeostasis
2. Concerning general anaesthesia, which of the following statements is true
   a) the triad of general anaesthesis is amnesia analgesia and muscle relaxation
   b) IV access must always be obtained prior to induction of anaesthesia
   c) Rapid sequence inductions are only indicating for unfasted patients undergoing non elective surgery
   d) nitrous oxide is a useful agent due to its analgesic effect and low risk or postoperative nausea and vomiting
3. Which of the following statements regarding burn depth is true
   a) Alkalis, including cement usually result in superficial burns
   b) fat burns are deeper than electrical contact burns
   c) the depth of burn together with the percentage of TBSA and smoke inhalation are key parameters in the assessment and management of a burn
   d) capillary filling is not present in superficial burns
4. Regarding history taking, which of the following is not routinely part of the process
   a) introduce yourself
   b) confirm the patient's name and date of birth
   c) obtain details of the patient's past medical history
   d) assess for tenderness
5. Which of the following descriptions of an ulcer edge denotes malignancy
   a) sloping
   b) overhanging
   c) everted and rolled
   d) punched out
6. Which of the following statements is true regarding chemical burns
   a) dry powder should be brushed away before irrigation with water
   b) hydrofluoric acid burns are associated with hypercalcaemia
   c) acid burns require longer irrigation than alkali burns
   d) neutralizing agents should be the first line of treatment in cement burns
7. In pleural space disease, which one of these statements is not an indication for surgical intervention in pneumothorax
   a) bilateral spontaneous pneumothoraces
   b) first ipsilateral spontaneous pneumothorax in a young tall male
   c) first contralateral spontaneous pneumothorax
   d) pneumothorax in professionals at risk like divers and pilots
8. Regarding orthopaedic imaging, which of the following statements is false
   a) x-ray is the first investigation in destructive bone lesions

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b) synovitis can be detected by plain x-ray
c) MRI arthrography is the ideal imaging for articular cartilage damage
d) MRI is the ideal method for local staging of a malignant bone tumour

9. Which of the following statements about mediators in metabolic response to trauma is true
   a) they consist of neural, endocrine and inflammatory
   b) they play an important role in flow phase
   c) every endocrine gland plays a part
   d) these mediators are released over several; days
10. The following statements are true regarding prophylaxis antibiotics except
   a) the blood and tissue levels of antibiotics should be maximal before contamination occurs
   b) in a long operation, under certain conditions antibiotics should be repeated at 4-hourly interval
   c) further doses of antibiotics postoperatively help in reducing infection
   d) patients with cardiac valve disease on a prosthesis should have prophylaxis
11. Patients vary greatly in their requirement for postoperative analgesia. What is the best way to assess adequacy of pain relief?
   a) measure the degree of tachycardia
   b) ask the patient to measure the pain
   c) assess the level of hypertension
   d) look for tachypneoa
12. A 50-year-old woman sustains a displaced and angulated fracture through the diaphysis of the right humerus. Following four months of treatment with initially a U slab then a humeral brace, the patient has persistent pain and mobility of the fracture site and check radiographs demonstrates no cortex bridging and no callus. What is the most likely diagnosis?
   a) Delayed union
   b) atrophic non union
   c) hypertrophic non union
   d) infected non union
13. Which of the following statements is true
   a) escharoctomy is not indicated for circumferential burns to the chest
   b) significant blood loss is not a feature to be considered when escharotomy is contemplated
   c) damage to major nerves can result by placing incorrect escharotomy incisions
   d) escharotomy of the hand and finger is best done outside of a main operating theatre
14. Regarding stress fractures, which of the following statements are true
   a) they most frequently present acutely following a high density high load sport
   b) pain is very well localized
   c) they are easy to diagnose with radiographs
   d) MRI is a useful adjustment in the diagnosis
15. Delayed wound healing following surgery may be caused by all the following except
   a) severe anaemia
   b) protein depletion

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c) good hydration status
d) cytotoxic drugs

16. The beck's triad of symptoms in cardiac tamponade include the following except
    a) increased jugular venous pressure
    b) hypotension
    c) hypertension
    d) muffled heart sounds
17. The following are early complications of burns except
    a) fluid and electrolyte loss
    b) death
    c) tetanus
    d) hypothermia
18. Regarding soft tissue injuries, which of the following statements is true
    a) damaged muscle can be replaced with cartilage
    b) meniscal tears of knees lead to rapid onset effusion
    c) soft tissue hematomas do not resolve spontaneously
    d) a cyst following soft tissue haematoma never requires surgical excision
19. Which of the following statements is not true in regard to breast abscess?
    a) Staphylococcus aureus is the most common
    b) Staphylococcus pyogens is the most common
    c) Lactation associated infections are most common
    d) It occurs normally within 6-8 weeks following delivery
20. All the following statements are surgical complications of obesity except
    a) cardiopulmonary complications such as cardiac failure
    b) wound complication such as infection, wound dehiscence
    c) venous thromboembolism such as pulmonary embolism
    d) none of the above statement is correct

SECTION 2: TRUE/FALSE QUESTIONS – (20 MARKS)

1. Which of the following statement regarding necrotizing fasciitis are true
    a) it is usually monomicrobial in majority of cases
    b) it always occurs in individuals with underlying immunological compromise T
    c) an absence of soft tissue gas in plain x-ray safety rules out this conditions F
    d) pain is the most important initial feature T
    e) surgery is reserved only if the condition doesn't improve with intravenous antibiotics for five days F
2. In Benign breast disease, the following statements are true
    a) nipple retraction always indicates an underlying malignancy
    b) blood stained nipple discharge is usually due to a breast carcinoma
    c) staphylococcus aureus is the most common cause for lactational mastitis

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d) fibroadenomas should always be offered excision as they are associated with a high risk malignancy
e) the histology of phyllodes tumour may be benign intermediate or malignant

3. Which of the following statements is true?
   a) Tension pneumothorax can cause haemodynamic compromise
   b) pleural effusions due to cardiac failure, renal failure, hepatic disease, inflammatory disease and malignancy have different protein content
   c) infection of the pleural space (empyema) results from iatrogenic and non-iatrogenic causes
   d) chest drains are no longer critical to the management of chest disease
   e) video assisted thoracoscopic surgery (VATS) plays a major role in the management of pleural space diseases
4. Regarding patients with diabetes undergoing surgery, which of the following statements are true
   a) poor wound healing is more common in diabetic patients
   b) diabetic patients should be prioritized on the theatre list
   c) HbA1C reflects glucose control over the preceding 8 to 12 weeks
   d) tight glucose control is recommended to prevent infection
   e) diabetic patients have an increased risk of cardiac events
5. Classical features of local anaesthetic toxicity include
   a) seizures
   b) loss of consciousness
   c) cardiac arrhythmias
   d) transient blindness
   e) hypertention
6. Which of the following answers regarding postoperative bleeding are true?
   a) post-operative bleeding can be difficult to diagnose
   b) postoperative bleeding is not significant unless Hb drops below 8g/dL
   c) patients should be resuscitated with warm fluids where possible
   d) postoperative bleeding is only significant in losses are over 1L
   e) patients should have a valid blood group and save prior to major surgery
7. For consent to be valid which of the statement are true
   a) the patient must be competent to give it
   b) the patient must not be coerced into decision making
   c) the patient must be given sufficient information
   d) if competence is severely compromised, the patient's care giver should assume responsibility
   e) is a patient has a legally valid advanced directive refusing treatment, the decision must be honoured
8. Which of the following statements are true regarding bacteria involved in surgical infection?
   a) Streptococci are most often responsible for cellulitis

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b) the most common organism in an abscess is staphylococci
c) bacterioides are aerobic spore-bearing organisms
clostridium difficile causes
e) staphylococcus epidermidis is a cause of hospital associated infection

9. Changes in body metabolism that occur in response to trauma are the following except
a) lipolysis
b) gluconeogenesis
c) protein breakdown
d) hypoglycaemia
e) hypermetabolism

10. Regarding preoperative management of patient with severe respiratory disease, which of the following statements are true
a) a course of antibiotics should always be given in patients with chronic sputum production
b) stopping smoking prior to surgery is of little benefit
c) regional anaesthesia should be considered where possible
d) a course of steroids prior to surgery may be necessary in patients with chronic obstructive airway disease
e) preoperative physiotherapy has no role

11. Which of the following is a risk factor for thrombosis
a) pregnancy
b) young age
c) smoking
d) trauma
e) malignancy

12. In regard to throracic injury, which of the following statements are true?
a) thoracic injury accounts for 25% of all severe injuries
b) an intercostal drain (ICD) is both diagnostic and therapeutic
c) CT scan is the most reliable investigation
d) 80% of chest injuries can be managed without an operation
e) in an open pneumothorax, close the pneumothorax and then insert intercostal drain (ICD)

13. Which of the following statements regarding burn complications are true?
a) cell-mediated immunity is increased in major burns
b) infections with bacteria and fungi are rare in large burns
c) malabsorption from gut damage is a known complication in a burns patient
d) circumferential full thickness burns of a limb can result in ischaemia
e) a change in voice is an important clinical sign in a burned patient

14. The following are recognized causes of avascular necrosis
a) alcohol excess
b) caisson disease
c) chronic liver disease
d) sickle cell disease

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e) smoking

15. Which of the following statements are true
    a) lymphedema in the arm after treatment usually indicates a recurrence
    b) radiotherapy after implant reconstruction of the breast gives poor cosmetic results
    c) chemotherapy is safe in second and third trimester in women with pregnancy associated breast cancer
    d) male breast cancer carries a poor prognosis when compare with female breast cancer
    e) none of the above is true
16. Regarding gangrene, which of the following statements are true
    a) wet gangrene occurs when there is infection
    b) neuropathy is an important factor in diabetic gangrene
    c) frostbite is gangrene caused by cold injury to the vessel wall
    d) bedsore is a type of gangrene
    e) amputation of the affected limb should be done in all cases at the earliest to prevent the spread of life-threatening sepsis
17. The following statements are the clinical tests of union following fracture healing
    a) absence of mobility between the fragments
    b) visible callus bridging the fracture and blending with both fragments
    c) absence of tenderness on firm palpation over the site of fracture
    d) continuity of bone trabeculae across the fracture
    e) absence of pain when angulation stress is applied at the site of fracture
18. Which of the following statements regarding spine injury are true
    a) all three columns of the spinal column must be injured for the spine to be unstable
    b) the cervico-thoracic junction is especially susceptible to injury because it is a transition zone from the mobile to the rigid segment of the spinal cord
    c) the size of the spinal canal in the cervical region makes the spine at this level especially susceptible to injury
    d) the spinal cord in an adult ends at L5/S1
    e) the stability of the cervical spine is mainly provided by the bony anatomy
19. The systemic early complication of fracture are
    a) deep venous thrombosis
    b) septicaemia in open fracture
    c) fat embolism syndrome
    d) disseminated intravascular coagulation (DIC)
    e) compartment syndrome
20. The following statements are risk factors that lead to formation of keloids in the body
    a) family history of keloids
    b) pregnancy
    c) adolescence
    d) certain locations on the body e.g. deltoids, earlobes, chest
    e) highly pigmented skin

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SECTION 3: SHORT ANSWER QUESTIONS – (40 MARKS)

1. Symptoms of a spinal fracture vary depending on the severity and location on the injury **(10 marks)**
   a) Name any four symptom that manifest with a spinal injury **(2 marks)**
   b) Diseases such as osteoporosis and spine tumors contribute to spine fractures. Name five primary causes of spinal fractures **(5 marks)**
   c) Mention any three radiological investigations that you would carry out in a spinal fracture **(3 marks)**
2. Tony Floyd, 25 years old present at casualty department with a history of a stab wound on the anterior right side of the chest following a scuffle in a pub. After a careful history taking and examination, the surgeon decides to put a chest tube drainage.
   a) What are the indications of a chest tube drainage in a chest injury **(5 marks)**
   b) What are the possible complications that may arise with insertion of the chest tube drainage **(5 marks)**
3. Miss P Anderson, 36 years old female underwent an emergency appendicectomy for an acute perforated appendicitis
   Outline the postoperative care that you will give to Miss P Anderson **(10 marks)**
4. Mr. Jafferson Hawkins works in Courier company as a motorcycle rider and gets involved in a road traffic accident and sustains a closed oblique fracture on the left mid humerus. The fracture is immobilized by a U slab. Explain the stages of bone repair that is likely to take place on the above fracture. **(10 marks)**
   (each point 2 marks with explanation)

SECTION 4: LONG ANSWER QUESTIONS – (20 marks)

1. Mr. T. James, 30 years' old who works with KPLC was electrocuted in the line of duty and sustained 2nd degree burns, approximately 50% TBSA. He was therefore referred to a burns centre
   a) Outline the emergency plan at the burns centre **(10 marks)**
   b) Name Burn Centre Referral criteria **(10 marks)**